

Report for EMIR-Specific RAG

1 Methodology

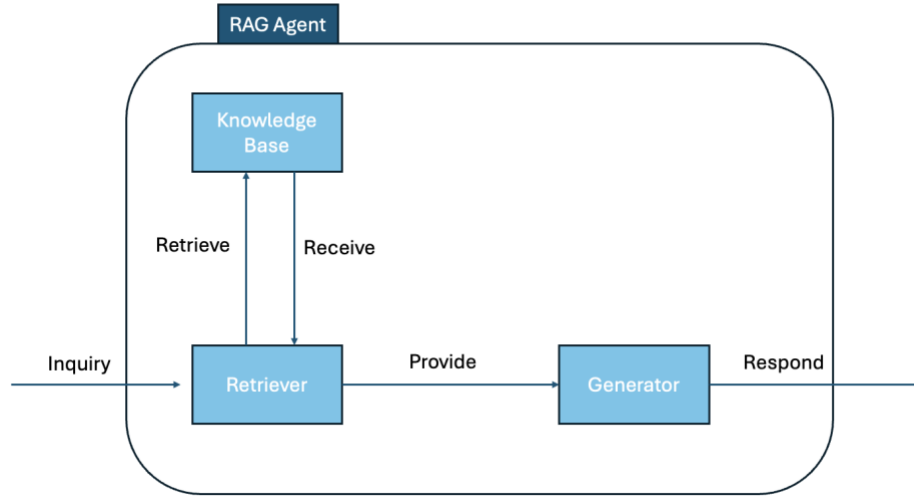


Figure 1. RAG System Framework

In our EMIR-specific Retrieval-Augmented Generation (RAG) system, the methodology is structured around three main components: the retriever, the knowledge base, and the generator. The retriever is responsible for transforming the user's question into embeddings using the all-MiniLM-L6-v2 model, and then searching the knowledge base to identify the most semantically similar sentences. The top matches are selected and passed along with the query to the generator, ensuring that the generated answer is grounded in relevant context. The knowledge base itself is composed of two complementary sources: (i) curated, well-structured offline data stored locally for accuracy and reliability, and (ii) online knowledge dynamically retrieved through APIs when the offline source lacks sufficient coverage. Finally, the generator—tested with LLaMA3 8B and Mistral 8B as baselines—receives both the query and the retrieved context to produce the final output. This pipeline design enables the system to combine domain-specific precision with the flexibility to incorporate external, up-to-date information.

2 Tasks and Results

For evaluation, we designed tasks based on three EMIR-related benchmarks, each reflecting different aspects of regulatory data: abbreviations, definitions, and U.S. financial laws and regulations. Our system was compared against the baseline setup of using the language models directly without retrieval augmentation. By conducting experiments on both settings, we were able to assess the contribution of the retriever–knowledge base pipeline to the overall answer quality. To further ensure reliability, we used GPT-4o as an external judge to verify whether the generated responses were correct and aligned with the benchmark ground truth. This combination of benchmark-based testing and LLM-based validation allowed us to quantitatively and qualitatively evaluate the improvements achieved by our EMIR-specific RAG system over baseline models.

Method/Model	Abbreviations	Definitions	U.S. Financial Laws	Avg. Accuracy
LLaMA3 8B (Base Model)	39.65%	56.00%	31.57%	42.41%
Mistral 8B (Base Model)	33.62%	50.00%	32.89%	38.84%
RAG (LLaMA3 8B)	51.72%	68.00%	35.52%	51.74%
RAG (Mistral 8B)	50.00%	58.00%	34.21%	47.40%

3 Evaluations

We evaluated our system on three EMIR-related benchmarks—Abbreviations, Definitions, and U.S. Financial Laws—comparing both base models with their corresponding RAG-enhanced versions. Overall, the RAG framework substantially improved accuracy. RAG (LLaMA3 8B) achieved the best average accuracy at 51.74%, representing an absolute improvement of +9.33 percentage points (a 22.0% relative gain) over the LLaMA3 8B baseline (42.41%). Similarly, RAG (Mistral 8B) reached 47.40%, improving by +8.56 percentage points (22.0% relative gain) compared to the Mistral 8B baseline (38.84%).

From a task-level perspective, the largest gains were observed in **Abbreviations** and **Definitions**: RAG (LLaMA3) improved by **+12.07** and **+12.00** points, while RAG (Mistral)

improved by **+16.38** and **+8.00** points, respectively. These results suggest that retrieval-augmented context significantly reduces hallucinations and improves precision when handling terminology and definition-based queries. In contrast, improvements on **U.S. Financial Laws** were more modest (RAG (LLaMA3): **+3.95**, RAG (Mistral): **+1.32**). This outcome can largely be attributed to limitations in the knowledge base, where coverage of relevant regulatory texts was insufficient, and the online search component was unable to consistently retrieve useful supporting content. As a result, the retriever provided little additional signal to the generator, leading to only marginal gains over the base models.

All outputs were independently validated using GPT-4o as an external judge, ensuring consistency and correctness relative to the benchmark ground truth. Taken together, these results demonstrate that the RAG architecture provides consistent and meaningful improvements over base models in the EMIR-specific setting, with the LLaMA3 8B + RAG combination yielding the strongest overall performance.

4 Conclusion

In this work, we developed an EMIR-specific Retrieval-Augmented Generation (RAG) system and demonstrated its effectiveness across multiple regulatory benchmarks. By integrating curated offline data with online search and leveraging a retriever-generator pipeline, our approach consistently outperformed base models, particularly in terminology- and definition-focused tasks. While gains on legal texts were more modest due to limited coverage in the knowledge base and challenges in retrieving high-quality online content, the overall results highlight the value of retrieval augmentation in improving accuracy and grounding responses in domain-relevant knowledge. Looking forward, expanding the legal corpus and refining retrieval strategies will be key to addressing these limitations. More broadly, our findings suggest that RAG architectures provide a promising pathway for building reliable domain-specific QA systems in financial regulation and beyond.